(e.g. “A Comparative Regression Study for Predicting <target variable>”)

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Abstract

This study investigates ...

Keywords: Regression, Machine learning, Predictive modeling,

1. Introduction

The remainder of this paper is organized as follows. Section 2 describes the dataset and methodology. Section 3 reports the experimental results. Section 4 discusses the findings and their limitations, and Section 5 concludes the paper.

2. Data and Methodology

2.1. Dataset

2.2. Preprocessing

2.3. Regression model(s)

$$\hat{y} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j + \varepsilon, \tag{1}$$

where $\hat{y}$ is the predicted target, x_j are the p predictors, β_j the estimated coefficients and ε the error term.

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2.4. Evaluation protocol

$$\mathrm{R}^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \qquad \mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|, \tag{2}$$

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}. \tag{3}$$

3. Results

3.1. Predictive performance

Table 1: Predictive performance of the evaluated regression models on the test set.

Model	R^2	MAE	MSE	RMSE
Model A	0.0000	0.0000	0.0000	0.0000
Model B	0.0000	0.0000	0.0000	0.0000

As reported in Table 1, the best-performing model is

3.2. Estimated coefficients

Table 2: Estimated coefficients of the <model name> model.

Predictor	Coefficient	Std. error	p-value
Intercept	0.0000	0.0000	0.0000
Feature 1	0.0000	0.0000	0.0000
Feature 2	0.0000	0.0000	0.0000

3.3. Diagnostic plots

Figure 1 shows . . . Figure 2 indicates . . .

4. Discussion

5. Conclusion and future work

fig_predicted_vs_actual.png

Figure 1: Predicted versus actual values for the <model name> model on the test set.

fig_residuals.png

Figure 2: Residual plot of the <model name> model. A random scatter around zero indicates that the model assumptions are reasonably satisfied.